PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERMENT

LONGIDO- DISTRICT COUNCIL

FORM THREE MID-TERM EXAMINATION

3 HRS

1.

I	II	III	IV	V	VI	VII	VIII	IX	X
D	C	C	A	A	A	A	D	D	D

1 @ Total = 10

2.

LIST A	I	II	III	IV	V
LIST B	H	A	G	C	E

5 Mark

3 (a) (i) Non-luminous flame used in the flame test of certain chemical substance.

(ii) Non-luminous flame used for heating purpose because it give a lot of heat.

(iii) Non-luminous flame suitable for cooking

1 @ total 3 mark

(b)

	Categorie	Examples
(i)	Apparatus for measuring	Pipette, Burette, and measuring cylinder
(ii)	Apparatus for testing	Test tube, Dropper and flask
(iii)	Apparatus for heating	Boiling tube, Burners, and Lamp
(iv)	Apparatus for filtering	Filter funnel, filter paper

1 @ Total = 4 marks

4. @ (i) K = 2:8:8:1 ½ Mark

L = 2:8:7 ½ Mark

(ii) L is not metal but is non-metal 1 mark

(iii) $K_1L_1 = KL$ 1 mark

b/ (a) Water
- Any type of portable fire extinguisher except carbondioxide

(b) Use fire blanket or sand if fire is small.

(c) Dry powder extinguisher

(d) Wet chemical extinguisher

1@ Total = 4 marks

5. (a) (i) Use of charcal lead to production of Carbonmonoxide which is poison gas.

(ii) Production of charcoal involve cutting down tree resulting to deforestation.

(iii) Due to excesive use of tree for making charcoal may lead to climatic change (drought)

(iv) Cause globe warming

1@ Total 3marks.

(b) (i) $Na_2SO_{4(aq)} + Ba(NO_3)_{2(aq)} \longrightarrow NaNO_{3(aq)} + BaSO_{4(s)}$ 1 mark

(ii) 1mol $NaSO_4$ ⟶ 1mol $BaSO_4$

3.5mol X

x?

$\frac{1mol\ x}{1mol} = \frac{3.5mol \times 1mol}{1mol}$

$x = 3.5mol.$

Mole of $BaSO_4 = 3.5mol.$ 2.5 marks

$BaSO_4 = 137 + 32 + (16 \times 4) = 233g/mol$

Molar mass of $BaSO_4 = 233g/mol$ 1 marks

From

$n = M/Molar\ mass$

mass of $BaSO_4 = 3.56mol \times 233g/mol$

$M = 829.48g$

Mass of $BaSO_4 = 829.48g$ 1.5 marks

6. $2KOH_{(aq)} + H_2SO_{4(aq)} \longrightarrow K_2SO_{4(aq)} + H_2O_{(l)}$

$2K^+_{aq} + 2OH^-_{(aq)} + 2H^+_{(aq)} + SO_4^{2-}{}_{(aq)} \longrightarrow 2K^+_{(aq)} + SO_4^{2-} + H_2O_{(l)}$

$H^+_{(aq)} + OH^-_{(aq)} \longrightarrow H_2O$ 1.5 Marks

(ii) $HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$

$H^+_{(aq)} + Cl^-_{(aq)} + Na^+_{(aq)} + OH^-_{(aq)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)} + H_2O_{(l)}$

$H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(l)}$. 1.5 Marks.

(b) ① Complete Information about reactants and products eg number of ~~molecules~~ moles

② Physical states of the substance

③ Necessary Condition for chemical equation eg Temperature Pressure and catalyst

④ Equation also show if the reaction is exothermic or endothermic

4 Marks

7. (a) Mole is amount of substance which Contain as many particulate entities (atom, ions, or molecules) as there are atoms in 12g of carbon-12. 2 Mark

(b) $KClO_{3(s)} \xrightarrow[\text{Heat}]{MnO_2} 2KCl_{(s)} + 3O_{2(g)}$ 01 Marks

$KClO_{3(s)} = 39.1 + 35.5 + (16 \times 3) = 122.5 g mol^{-1}$

Molar mass of $KClO_3$ = 122.5 g/mol

$n = \frac{\text{Mass}}{\text{Molar mass}}$

$n = \frac{20g}{122.5 g/mol} = 0.163 mol$ 02 Marks

Number of mole of $KClO_3$ = 0.163

1mol of $KClO_3$ Produce 3mol of O_2

1mol → 3mol

0.163 → x

$x = 0.489$ mol. 2 marks

1mol → $22.4 dm^3$

0.489mol → ? x

$x = 0.489 \times 22.4 dm^3$

$x = 10.95 dm^3$

Volume of O_2 = $10.95 dm^3$

8. (a) (i) Isotopes are the atoms with same atomic number but different mass number.

2 Marks

WHILE

Isotopy is the ~~exi~~ tendency of atom of the same element to have same atomic number but different mass number

(ii) Ion is the charge element WHILE Atom is the small Particles that made atom.

(b). (i) Atom can be created or destroyed or spliat by mean of nuclear reaction.

(ii) Some element have atom of more than one kind which differ slightly in mass

(iii) A atom is made up of similar sub-atomic particle called Protons, Neutrons and electrons.

(iv) Atoms of different element may combine in many different ratio to form Complex compound.

10@ Total 4 marks

9. (a) (i) Ammonium Carbonate
(ii) Dinitrogen Pentoxide
(iii) Lead II oxide

10@ Total = 3 Marks

(b)

Element	C	S
% Composition	15.8%	84.2%
R.A.M	12	32
$\frac{\%\ \text{Composition}}{\text{RAM}}$	$\frac{15.8}{12}$	$\frac{84.2}{32}$
Value obtained divide by small value	$\frac{1.32}{1.32}$	$\frac{2.23}{1.32}$
Value obtained	1	2

Empirical formula = CS_2 —— 2.5 Marks

Molecular formular = n x empirical formula

But R.M.M = n x (R.A.M sum)

$76 = n \times [12 + (2 \times 32)]$

$\frac{76}{76} = \frac{76n}{76}$

∴ Molecular fomula = CS_2 2.5 Marks

10. (i) Is the solution with known Concentration. 1 marks

(ii) Is the solution which Contain one mole of solute dissolved to make one litre $1dm^3$ of solution. 1 mark

(iii) Is the quantity of solute present In given quantity of solution, 1 Marks

(b) (i) Data

Concentration = $8.6 g/dm^3$

Molarity of acid = 0.2M

Mole ratio = 1:1

Volume of acid (Va) = $12.5cm^3$

Volume of base (Vb) = $25cm^3$

Mole of acid (n_a) = 1

Mole of base (n_b) = 1.

$$M_b = \frac{M_a V_a n_b}{V_b n_a}$$

$$M_b = \frac{0.2M \times 12.5cm \times 1mol}{25cm^3 \times 1mol}$$

$M_b = 0.1M$ ——— 2.5 Marks

From Molarity = $\frac{\text{Concentration}}{\text{Molar mass}}$

$$0.1 = \frac{8.4}{\text{Molar mass}}$$

$$\text{Molar mass} = \frac{8.4}{0.1} = 84$$ 2.5 Marks

Hence $23 + x + 12 + 48 = 84$

$x = 84 - 83$

$x = 1$

2 Marks

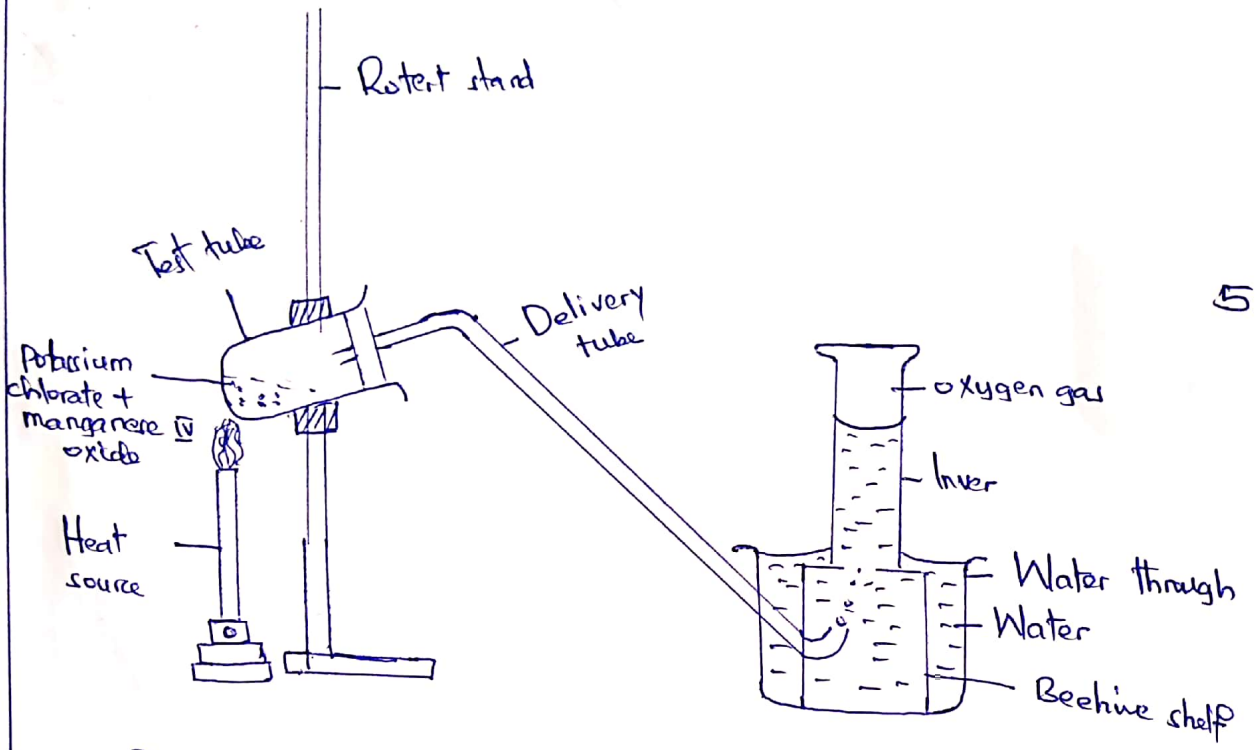
(ii) 1 1 Marks

(iii) Hydrogen 2 Marks

(iv) Hydrogen Carbonate 1 mark

(v) Negative one (-) 1 mark

11. (a)



5 Marks

(i) Potassium chlorate is decomposed by heating in the presence of manganese IV oxide catalyst to produce potassium chloride and oxygen gas. 2 Marks

(ii) $2KClO_{3(s)} \xrightarrow[\text{Heat}]{MnO_2} 2KCl_{(s)} + 3O_{2(g)}$ 2 Mark

(iii) So as to increase the rate of decomposition of potassium chlorate 1 Mark

(iv) (a) It is slightly soluble in water 1 Mark.

(b) It is lighter than water 1 mark.

(b) (i) Because hydrogen is light gas 1 mark

~~(ii) Because hydrogen is used in synthesis of ammonia by reacting it with~~

(ii) Because hydrogen is more reactive 1 Mark

(iii) Highly flammable 1 Mark